

CTF Challenge Walkthrough

Base64 Cryptography Challenge

Challenge Overview

Challenge Name	Base64 CTF Challenge
Category	Cryptography / Web Enumeration
Difficulty	Beginner
Flag Format	FLAG{...}
VM Login	ctfuser / ctfuser
Target IP	172.21.122.151

Tools Used

VirtualBox	Run the CTF Virtual Machine
Kali Linux	Attacking machine
Nmap	Discover IP address and open ports
Gobuster	Brute force hidden files on web server
curl	Retrieve the encoded flag from server
base64	Decode the flag to plaintext

PART 1 — Setup Side (Challenge Creator)

Step 1 — Install Ubuntu Server on VirtualBox

Created a new VM with 2GB RAM and 20GB disk. Installed Ubuntu Server 22.04 LTS with user ctfuser / ctfuser.

Step 2 — Encode the Flag

Encode the flag using Base64:

```
echo -n "FLAG{b4su64_1s_fun}" | base64
```

Output: RkxBR3tiNHN1NjRfMXNfZnVufQ==

Step 3 — Install Apache Web Server

```
apt install apache2 -y
```

Step 4 — Plant the Encoded Flag

```
echo "RkxBR3tiNHN1NjRfMXNfZnVufQ==" > /var/www/html/secret.txt
```

Step 5 — Create Fake Homepage

```
echo "Welcome to the server." > /var/www/html/index.html
```

Step 6 — Start Apache

```
systemctl start apache2  
systemctl enable apache2
```

Step 7 — Export VM as OVA

VirtualBox -> File -> Export Appliance -> CTF-Challenge.ova

PART 2 — Solver Side (Students)

Step 1 — Import the OVA

VirtualBox -> File -> Import Appliance -> Select CTF-Challenge.ova -> Import -> Start

Step 2 — Discover Target IP With Nmap

```
nmap -sn 172.21.122.0/24
```

Output: Nmap scan report for 172.21.122.151 — Host is up

Step 3 — Scan Open Ports

```
nmap 172.21.122.151
```

Output: 80/tcp open http — Web server is running!

Step 4 — Visit the Website

```
curl http://172.21.122.151
```

Output: Welcome to the server. (Nothing useful — keep searching)

Step 5 — Brute Force Hidden Files With Gobuster

```
gobuster dir -u http://172.21.122.151 -w /usr/share/wordlists/dirb/common.txt -x txt
```

Output: /secret.txt (Status: 200) — Hidden file discovered!

Step 6 — Retrieve the Encoded Flag

```
curl http://172.21.122.151/secret.txt
```

Output: RkxBR3tiNHN1NjRfMXNfZnVufQ== (The == at the end means Base64!)

Step 7 — Decode the Flag

```
echo "RkxBR3tiNHN1NjRfMXNfZnVufQ==" | base64 -d
```

Output: FLAG{b4su64_1s_fun} — Challenge Complete!

Full Attack Flow

#	Command	Result
1	nmap -sn 172.21.122.0/24	Found target: 172.21.122.151
2	nmap 172.21.122.151	Port 80 open — Web server running
3	curl http://172.21.122.151	Fake homepage — nothing useful
4	gobuster dir ... -x txt	Found /secret.txt (Status 200)
5	curl .../secret.txt	Got encoded string RkxBR3t...
6	echo 'RkxBR3t...' base64 -d	FLAG{b4su64_1s_fun}

Key Concepts Learned

Base64 Encoding	Encodes binary data as text using 64 characters. NOT encryption — anyone can decode it in
Port Scanning	Using nmap to discover what services are running on a target machine.
Directory Bruteforce	Using gobuster to find hidden files and directories on a web server.
Web Enumeration	Exploring a web server methodically to find hidden or sensitive content.

Lesson Learned

Base64 is NOT encryption. It is simply encoding. Anyone who recognizes it can instantly decode it using a single Linux command or an online tool. Never use Base64 to protect sensitive data.

FLAG{b4su64_1s_fun}
